

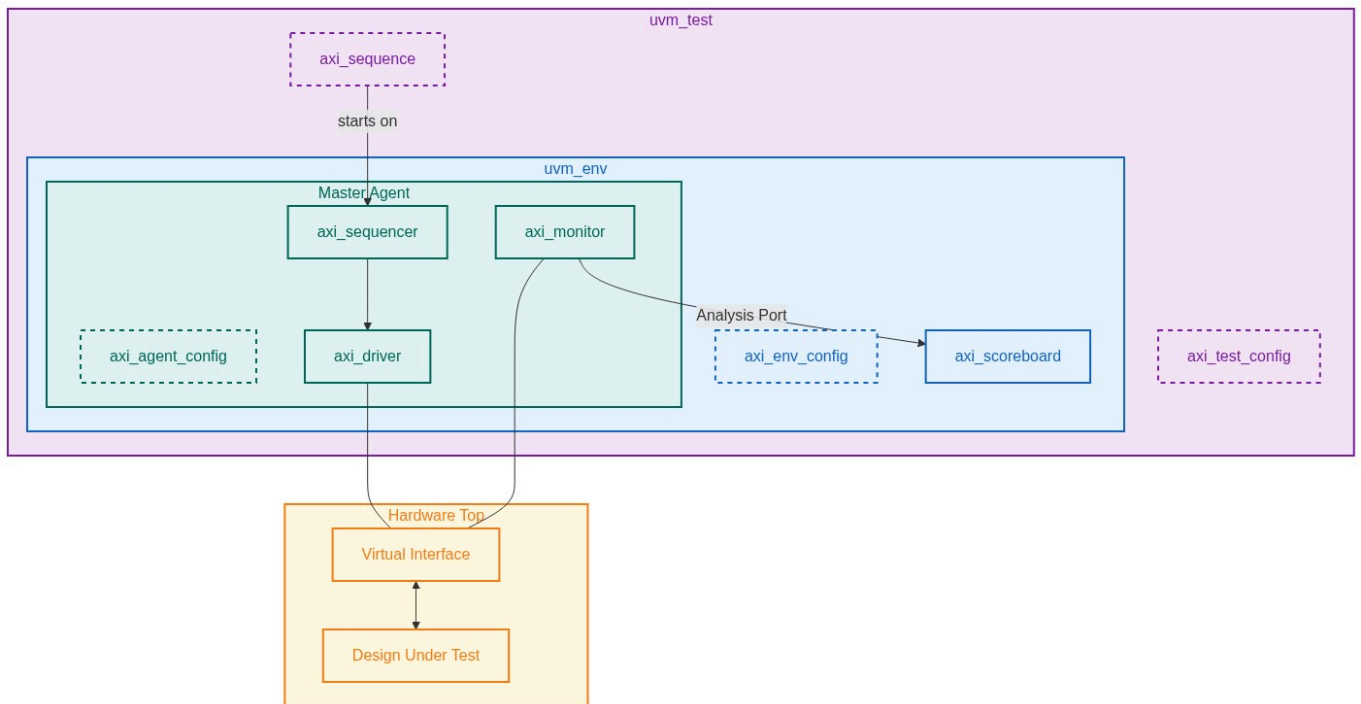
Verification Plan Report

Project: AXI-Stream Slave Functional Verification Plan

Version: 4.0

Verification Role: Slave

1. UVM Environment Architecture



2. Protocol Configurations

Feature	Configuration Name	Impact
Handshake Protocol	TVALID/TREADY Flow Control	Impacts the responsiveness of the slave agent and the driver's ability to inject backpressure.
Packet Framing	TLAST Signal Connectivity	Determines how the scoreboard reconstructs packets and detects boundary violations.
Data Integrity	Odd Parity (AXI5)	Requires parity checkers in the monitor and scoreboard to verify the Master's generation logic.

3. Test Requirements

Index	Name	Description
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Verification Plan Report

REQ_FLOW_CONTROL	Two-way Flow Control Stability	The Transmitter (Master Agent) must maintain TVALID high until the Slave (DUT) asserts TREADY. Verification role: Ensure the Slave DUT correctly handles stalls without skipping beats.
REQ_PARITY_CHECK	Odd Parity Verification	Odd parity across TDATACHK bits covering the payload. Verification role: Monitor checks the incoming stream for parity errors generated by the Master.

4. Test Cases & Mapping

Index	Scenario	TR Mapping
TC_SLAVE_READ_Y_LATENCY	Slave (DUT) asserts TREADY after N cycles of VALID; verify data is not sampled prematurely.	REQ_FLOW_CONTROL
TC_PARITY_FAILURE_DETECTION	Agent injects parity error on TDATACHK; verify DUT triggers error reporting.	REQ_PARITY_CHECK

5. Functional Coverage Groups

Group Name	Description
cg_ready_logic	Cover TREADY assertion latency from 0 to 16 cycles.

6. Interface Signals

Signal Name	Direction	Width	Description
TVALID	Input	1	Master-driven flow control.
TREADY	Output	1	Slave-driven flow control.

7. Verification Environment Structure

Root: `axi_stream_slave_verification/`

- |-- agents : Master Agent for Slave DUT
- |-- env : Environment with Scoreboard
- |-- verif_plan : Configuration and planning artifacts